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# NPS/001/010 – Technical Specification for Fasteners and Fixings for Wood Pole Overhead Lines and General Construction Works

## 1. Purpose

The purpose of this document is to detail the requirements for all fasteners and fixings to be utilised on the CE Electric UK distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

| Ref         | Version | Title                                                                                                          |
|-------------|---------|----------------------------------------------------------------------------------------------------------------|
| MS-O-15     | 1.0     | NEDL Specification for Fasteners and Fixings for Wood Pole Overhead Lines and General Construction Works       |
| NPS/001/010 | 1.0     | Technical Specification for Fasteners and Fixings for Wood Pole Overhead Lines and General Construction Works. |
| NPS/001/010 | 2.0     | Technical Specification for Fasteners and Fixings for Wood Pole Overhead Lines and General Construction Works. |

## 2. Scope

This document applies to the materials, dimensions and finish of bolts, eye bolts, eye nuts, nuts, screws, tie-rods, wall fixings and washers used for general construction works on the CE Electric UK distribution network.

This document includes details of national, international and industry standards that are listed within the technical particulars for each fastener type.

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## 2.1 Contents

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## 3. Technical Requirements

### 3.1 General Requirements

Fastenings and fixings will generally be in compliance with the requirements of ENA TS 43-96, Issue 2 - 1986. A self certification conformance declaration has been included in Appendix 1 and shall be completed by the supplier. A schedule of items required is listed in Appendix 2.

Where the specified item consists of multiple component parts they shall be supplied pre-assembled e.g. earth bolts comprise of a bolt, two half nut and two washers

### 3.2 Bolts and Nuts

#### 3.2.1 General

Bolts and nuts shall be ISO metric black hexagonal to BS 4190: 2000 and galvanised to BS EN 1461: 2009. The strength grade for bolts shall be grade 4.6 and grade 4.0 for nuts.

#### 3.2.2 Pole and Assembly Bolts

Pole and assembly bolts shall comply with ENA TS 43-96, Issue 2 – 1986, Tables 2 and 3 and be supplied complete with one full nut.

#### 3.2.3 Nuts

Nuts shall normally be supplied as full nuts unless otherwise stated.

#### 3.2.4 Earth Bolts

Steel earth bolts shall be in accordance with BS 4190: 2000 Form E and F and supplied with two half nuts and two round steel washers.

#### 3.2.5 Eye Bolts and Eye Nuts

Eye Bolts shall be manufactured in accordance with ENA TS 43-96, Issue 2 - 1986 Fig. 2 and supplied complete with one full nut. They shall preferably be of drop forge manufacture. Eye nuts shall be in accordance with ENA TS 43-96, Issue 2 - 1986 Fig. 1 and have metric, course series, screw threads complying with BS 3643 Parts 1 and 2: 2007.

#### 3.2.6 Tie Bolts

Tie bolts shall be in accordance with ENA TS 43-96, Issue 2 - 1986 Fig. 5 and supplied with four full nuts.

### 3.3 Washers

#### 3.3.1 Round Flat Washers

Round flat washers shall be manufactured from mild steel to BS 4320: 1968 Table 3, Form E and suitable to accommodate M10, M12, M16, M20 and M24 bolts as detailed in ENA TS 43-96, Issue 2 - 1986 Table 6.

#### 3.3.2 Square Flat Washers

Square flat washers shall be manufactured from BS 4360 grade EN 10025 Fe430a steel in accordance with ENA TS 43-96, Issue 2 - 1986 Table 6.

#### 3.3.3 Tapered Washers

Tapered washers shall be 'D' shaped and manufactured from grade BS EN 10 025 Fe430a steel with dimensions and design in accordance with ENA TS 43-96, Issue 2 - 1986.

#### 3.3.4 Square Curved Washers

Square curved washers shall be manufactured from grade BS EN 10 025 Fe430a steel in accordance with ENA TS 43-96, Issue 2 - 1986 Fig. 3.

|                           |     |                        |          |             |   |           |    |
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### 3.4 Coach Screws

Coach screws shall be manufactured from galvanised heat treated steel with a minimum strength of 40 kg/mm<sup>2</sup>, gimlet pointed in accordance with ENA TS 43-96, Issue 2 – 1986 Fig. 6, Drawing number 439609.

The thread form shall be uniform over the length except where the thread blends with the shank and point. The thread depth shall be 15% of the nominal screw diameter +/- 0.1mm. The number of threads shall be in accordance with the table in clause 7.1 of ENA TS 43-96, Issue 2 – 1986.

The point of the screw shall be tapered over a length 2 x the nominal diameter or  $\frac{2}{3}$  of the nominal screw length, whichever is the least. The taper from the shank to the root of the thread shall be approximately one half thread. The remaining portion of the screw shall be parallel.

### 3.5 Electroplated Screw Sets, Nuts and Washers

Screw sets, nuts and washers shall be manufactured from Grade 8.8 high tensile steel and electroplated in accordance with BS EN 4042: 2000. Washers and spring washers shall be supplied to accommodate M6, M8, M10, M12 and M16 bolts.

### 3.6 Roundhead Drive Screws

Roundhead drive screws shall be of galvanised steel (40kgf/mm<sup>2</sup>) of a form in accordance with BS 1494, Part 1, 1964: 6.0mm diameter and 65mm long as detailed in table 6 of ENA TS 43-96, Issue 2 - 1986.

### 3.7 Wood Screws

Wood screws shall be manufactured in accordance with BS 1210; 1983 – Specification for wood screws. They shall be supplied as specified in the product description as shown in Appendix 2 – Schedule of Items with options for round head or countersunk with a slotted or pozi-drive head.

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## 4. References

### 4.1 External Documentation

| Reference                   | Title                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| ENA TS 43-96 Issue 2: 1986  | Fasteners and Washers for Wood Pole Overhead Lines                                                              |
| BS EN ISO 1461 - 2009       | Hot Dipped Galvanised Coating on Fabricated Iron or Steel Articles                                              |
| BS 4190 - 2000              | Specification for ISO Metric Black Hexagonal Bolts, Screws and Nuts                                             |
| BS 3643 Part 1 and 2 - 2007 | ISO Metric Screw Threads                                                                                        |
| BS 4320 - 1968              | Specification for Metal Washers for General Engineering Purposes, Metric Series                                 |
| BS 4360 - 1968              | Specification of Weldable Structure Steel.<br>Note: to be read in association with BS EN 10 1025                |
| BS EN 10 025 - 2004         | Specification for Hot Rolled Products of Non-alloy Structural Steels and their Technical Delivery Conditions    |
| BS 1210 - 1963              | Specification for Wood Screws                                                                                   |
| BS EN ISO 4042: 2000        | Specification for Electroplated Coatings                                                                        |
| BS 1494 Part 1 - 1964       | Specification for Fixing Accessories for Building Purposes: Part 1 – Fixings for Sheet, Roof and Wall Coverings |

### 4.2 Internal documentation

| Reference | Title |
|-----------|-------|
|-----------|-------|

### 4.3 Amendments from Previous Version

| Clause/Section | Subject           | Amendments        |
|----------------|-------------------|-------------------|
| Appendix 2     | Schedule of items | Added to document |

## 5. Definitions

| Term | Definition |
|------|------------|
|------|------------|

|                           |     |                        |          |             |   |           |    |  |
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## 6. Authority for issue

|                                |                  |                                  |            |                  |
|--------------------------------|------------------|----------------------------------|------------|------------------|
| Written by:                    | Steven Salkeld   | Standards Engineer               | 11/01/2010 | Steven Salkeld   |
| Technically assured by:        | Richard McMahon  | Standards Engineer               | 12/01/2010 | Richard McMahon  |
| CDS assured by:                | Sean Johnson     | CDS Administrator                | 22/01/2010 | Sean Johnson     |
| Authorised for Publication by: | Chris Holdsworth | Standards and Assessment Manager | 18/01/2010 | Chris Holdsworth |

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## Appendix 1 - SELF CERTIFICATION CONFORMANCE DECLARATION

Fasteners and fixings shall comply with the latest issues of the relevant international and international standards.

ENATS 43-96 Issue 3: 1995 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses of the aforementioned standard relevant to fasteners and fixings for use on the CE Electric UK distribution network.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

### Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

### Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

**Manufacturer:**

**Product References:**

**Name:**

**Signature:**

**Date:**

|                           |     |                        |          |             |   |           |    |
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## Appendix 1

### Fasteners and Fixings for Wood Pole Overhead Lines and General Construction Works

| • ENATS 43-96 Issue 3: 1995 |                                                      |                  |         |
|-----------------------------|------------------------------------------------------|------------------|---------|
| Clause/Sub-clause           | Requirement                                          | Conformance Code | Remarks |
| 3.0                         | Materials and finish                                 |                  |         |
| 4.1                         | Technical particulars including “safe working loads” |                  |         |
| 4.4                         | Eye bolts and nuts including “safe working loads”    |                  |         |

|                           |             |                        |          |             |   |           |    |
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## Appendix 2 –Schedule of Items

| Item   | Description                                                                                          |
|--------|------------------------------------------------------------------------------------------------------|
| 265628 | ABC Fit Hook Bolt Extended 320mm,                                                                    |
| 265632 | ABC Fit Hook Nut M20                                                                                 |
| 265613 | ABC Fit Pig Tail Hook Bolt Extended M20 240mm,                                                       |
| 265609 | ABC Fit Pig Tail Hook Bolt M20, 320mm                                                                |
| 265596 | ABC-Fit M20 Pigtail Hook Bolt 220MM                                                                  |
| 371210 | Anchor Bolt, M12 x 100mm, Hilti Fixing                                                               |
| 374929 | Bolt, Earthing M12 x 25mm Brass, 25mm Thread, Complete with Two Round Washers and Two Half Nuts      |
| 370491 | Bolt, Earthing M12 x 40mm Galvanised, 30mm Thread, Complete with Two Round Washers and Two Half Nuts |
| 370487 | Bolt, Earthing M12 x 55mm Galvanised, 30mm Thread, Complete with Two Round Washers and Two Half Nuts |
| 370504 | Bolt, Earthing M20 x 60mm Galvanised, 46mm Thread, Complete with Two Round Washers and Two Half Nuts |
| 218115 | Bolt, M10 x 130mm Galvanised, 60mm Thread, Complete with Full Nut                                    |
| 378417 | Bolt, M10 x 50mm Galvanised, 40mm Thread, Complete with Full Nut                                     |
| 378474 | Bolt, M12 x 100mm Galvanised, 30mm Thread, Complete with Full Nut                                    |
| 218131 | Bolt, M12 x 25mm Galvanised, 15mm Thread, Complete with Full Nut                                     |
| 378440 | Bolt, M12 x 75mm Galvanised, 30mm Thread, Complete with Full Nut                                     |
| 378629 | Bolt, M16 x 220 Galvanised, 150mm Thread, Complete with Full Nut.                                    |
| 378633 | Bolt, M16 x 260 Galvanised, 150mm Thread, Complete with Full Nut                                     |
| 378652 | Bolt, M16 x 300 Galvanised, 150mm Thread, Complete with Full Nut                                     |
| 378671 | Bolt, M16 x 400mm Galvanised, 150mm Thread, Complete with Full Nut.                                  |
| 378101 | Bolt, M20 x 140mm Galvanised, 100mm Thread, Complete with Full Nut.                                  |
| 378794 | Bolt, M20 x 300 Galvanised, 150mm Thread, Complete with Full Nut                                     |
| 378794 | Bolt, M20 x 300 Galvanised, 150mm Thread, Complete with Full Nut                                     |
| 372355 | Bolt, M20 x 600 Galvanised, 200mm Thread, Complete with Full Nut                                     |

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| 372552 | Bolt, M22 x 70mm Galvanised, 50mm Thread, Complete with Full Nut.      |
| 378421 | Bolt: M12 x 40mm & 30mm Thread: Galvanised & Complete With Full Nut.   |
| 378436 | Bolt: M12 x 40mm & 30mm Thread: Galvanised & Complete With Full Nut.   |
| 378436 | Bolt: M12 x 40mm & 30mm Thread: Galvanised & Complete With Full Nut.   |
| 378582 | Bolt: M16 x 130mm & 100mm Thread: Galvanised & Complete With Full Nut. |
| 378544 | Bolt: M16 x 50mm & 38mm Thread: Galvanised & Complete With Full Nut.   |
| 378756 | Bolt: M20 x 220mm & 150mm Thread: Galvanised & Complete With Full Nut. |
| 372904 | Bolt: M20 x 400mm & 150mm Thread: Galvanised & Complete With Full Nut. |
| 372815 | Bolt: M20 x 530mm & 150mm Thread: Galvanised & Complete With Full Nut. |
| 378703 | Bolt: M20 x 60mm & 46mm Thread: Galvanised & Complete With Full Nut.   |
| 373911 | Bolt: M20 x 70mm & 46mm Thread: Galvanised & Complete With Full Nut.   |
| 297812 | Coach Screw, M10 x 120 Galvanised,                                     |
| 297754 | Coach Screw, M10 x 50 Galvanised                                       |
| 297986 | Coach Screw, M10 x 75 Galvanised                                       |
| 297762 | Coach Screw, M10 x 90 Galvanised.                                      |
| 372919 | Coach Screw, M12 x 110 Galvanised.                                     |
| 297770 | Coach Screw, M12 x 75 Galvanised.                                      |
| 297721 | Coach Screw: M6 x 50mm: Galvanised.                                    |
| 368082 | Eyebolt, M20 x 100mm Galvanised, Complete with Full Nut                |
| 368044 | Eyebolt, M20 x 200mm Galvanised, Complete with Full Nut                |
| 368059 | Eyebolt, M20 x 300mm Galvanised, Complete with Full Nut                |
| 250270 | Eyebolt, M20 x 400mm Galvanised, Complete with Full Nut                |
| 369507 | Full Nut M10 Electroplated.                                            |
| 369598 | Full Nut M12 Electroplated.                                            |
| 369691 | Full Nut M16 Electroplated.                                            |
| 369441 | Full Nut M6 Electroplated                                              |
| 369456 | Full Nut M8 Electroplated.                                             |
| 378883 | Full Nut, M10 Galvanised                                               |
| 379405 | Full Nut, M22 Galvanised                                               |
| 378898 | Full Nut: M12: Galvanised.                                             |
| 378915 | Full Nut: M20 Galvanised.                                              |

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|        |                                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 371653 | Nails, Oval: 3.0": (per kg)                                                                                                                                                              |
| 371579 | Nails, Round: 4.0" X 9 Gauge: (per kg)                                                                                                                                                   |
| 371526 | Nails, Round: 1.5" x 12 Gauge. (per kg)                                                                                                                                                  |
| 371564 | Nails, Round: 2.5" X 11 Gauge: (per kg)                                                                                                                                                  |
| 378402 | Nails: Roofing: M6 x 65mm: Galvanised: Convex Head: (per kg)                                                                                                                             |
| 371244 | Nails; Round Plain head 1.8 x 25mm Galvanised to BS 1202 Part 1, Table 1 (in packs of 200)                                                                                               |
| 371140 | Rawl Through bolt: 10mm X 95mm                                                                                                                                                           |
| 254300 | Rawl Through bolt: M10 x 30mm                                                                                                                                                            |
| 254292 | Rawl Through bolt: M10 x15mm x 75mm                                                                                                                                                      |
| 371155 | Rawl Through bolt: M12 x 135mm.                                                                                                                                                          |
| 254284 | Rawl Through bolt: M8 x 25mm                                                                                                                                                             |
| 371738 | Screw Nail: 50mm: BZP: Flat Head: (Box of 500)                                                                                                                                           |
| 369579 | Screw Set: Electroplated M6 x 25mm.                                                                                                                                                      |
| 425827 | Screw Set: Electroplated: M10 x 25mm: Hex Head.                                                                                                                                          |
| 425835 | Screw Set: Electroplated: M10 x 40mm: Hex Head.                                                                                                                                          |
| 425876 | Screw Set: Electroplated: M12 x 40mm: Hex Head.                                                                                                                                          |
| 425884 | Screw Set: Electroplated: M12 x 50mm: Hex Head.                                                                                                                                          |
| 379212 | Screw Set: Electroplated: M12 x 60mm: Hex Head.                                                                                                                                          |
| 425785 | Screw Set: Electroplated: M8 x 25mm: Hex Head.                                                                                                                                           |
| 425819 | Screw Set: Electroplated: M8 x 50mm; Hex Head.                                                                                                                                           |
| 350500 | Spring - stainless steel spring 5mm i/d 8mm o/d 80mm long spring to be "weak". (to increase diameter of thin shackle lock when used on aerial switches. must easily thread onto shackle. |
| 299214 | Staple 1.1/2" Galvanised 8SWg (per kg)                                                                                                                                                   |
| 299222 | Staple, Galvanised 60mm x 6mm x 21mm                                                                                                                                                     |
| 374416 | Staple: Copper: No12 x 1"                                                                                                                                                                |
| 374399 | Staple: Copper: No8 x 1.5"                                                                                                                                                               |
| 375599 | Staple: M9 x 1-1/4": Galvanised: (per kg)                                                                                                                                                |
| 372730 | Studding, Steel Zinc Electroplated M10 x 300mm Grade 4.6, Table 14                                                                                                                       |
| 372745 | Studding, Steel Zinc Electroplated M12 x 300mm Grade 4.6, Table 14                                                                                                                       |
| 372726 | Studding, Steel Zinc Electroplated M8 x 300mm Grade 4.6, Table 14                                                                                                                        |
| 375828 | Tie Rod, M20 x 330 Galvanised Complete with Four Full Nuts                                                                                                                               |
| 375832 | Tie Rod, M20 x 450 Galvanised Complete with Four Full Nuts                                                                                                                               |
| 375902 | Tie Rod, M20 x 550 Galvanised Complete with Four Full Nuts.                                                                                                                              |
| 371969 | Wall Plug: Plastic - Brown                                                                                                                                                               |
| 371954 | Wall Plug: Plastic - Red.                                                                                                                                                                |
| 371988 | Wall Plug: Plastic: M12 x 50mm Grey or Green                                                                                                                                             |
| 369668 | Washer M10 Electroplated.                                                                                                                                                                |

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| 369672 | Washer M12 Electroplated.                                                 |
| 369437 | Washer M6 Electroplated.                                                  |
| 379852 | Washer, Galvanised M16 Tapered - 'D' Shaped                               |
| 374488 | Washer, Galvanised M20 Tapered - 'D' Shaped                               |
| 350799 | Washer: Bellville M12: Standard Finish.                                   |
| 353301 | Washer: Bellville M16: Standard Finish.                                   |
| 352135 | Washer: Bellville M8: Standard Finish.                                    |
| 351715 | Washer: Bellville: M20 Standard Finish.                                   |
| 351912 | Washer: Bellville: M10 Standard Finish.                                   |
| 375601 | Washer: M12: Galvanised.                                                  |
| 375601 | Washer: M12: Galvanised.                                                  |
| 369494 | Washer: M16 Electroplated.                                                |
| 378987 | Washer: M16: Galvanised.                                                  |
| 378987 | Washer: M16: Galvanised.                                                  |
| 368078 | Washer: M20 Square Curved, 65mm x 65mm.                                   |
| 368063 | Washer: M20, Flat Square 65mm x 65mm.                                     |
| 375616 | Washer: M20: Galvanised.                                                  |
| 369653 | Washer: M8 Electroplated.                                                 |
| 369615 | Washer: Spring: M10: Electroplated.                                       |
| 369634 | Washer: Spring: M12: Electroplated.                                       |
| 369475 | Washer: Spring: M8: Electroplated.                                        |
| 229450 | Washer Galvanised MS Round (Flat) to suit M10 Bolt                        |
| 372158 | Wood Screw: No 10 x 1": Steel Countersunk: Pozi-Drive. (Box of 200)       |
| 372162 | Wood Screw: No 10 x 1-1/2": Steel Countersunk: Pozi-Drive. (Box of 200)   |
| 376958 | Wood Screw: No 10 x 1-1/4": Steel Countersunk: Pozi-Drive. (Box of 200)   |
| 372177 | Wood Screw: No 10 x 2": Steel Countersunk: Pozi-Drive: (Box of 200)       |
| 376962 | Wood Screw: No 10 x 2-1/2": Steel Countersunk: Pozi-Drive. (Box of 200)   |
| 372181 | Wood Screw: No 10 x 3": Steel Countersunk: Pozi-Drive. (Box of 200)       |
| 372209 | Wood Screw: No 12 x 1-1/2": Steel: Countersunk: Pozi-Drive: (Box of 200)  |
| 426155 | Wood Screw: No 12 x 2 1/2": Brass: Countersunk: (Box of 200)              |
| 372228 | Wood Screw: No 12 x 2-1/2": Steel: Countersunk: Pozi Drive: ( Box of 200) |
| 372497 | Wood Screw: No 6 x 1": Brass Round Head: (Box of 200)                     |
| 426288 | Wood Screw: No 8 x 1 1/2": Brass Round Head: (Box of 200)                 |

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| 375777 | Wood Screw: No 8 x 1": Brass Round Head: (Box of 200).                 |
| 372073 | Wood Screw: No 8 x 1": Steel Countersunk: Pozi-Drive. (Box of 200)     |
| 372088 | Wood Screw: No 8 x 1-1/4" Steel Countersunk: Pozi-Drive. (Box of 200)  |
| 426296 | Wood Screw: No 8 x 2": Brass Roundhead: (Box of 200)                   |
| 372105 | Wood Screw: No 8 x 2": Steel Countersunk: Pozi-Drive. (Box of 200)     |
| 372124 | Wood Screw: No 8 x 2-1/2"; Steel Countersunk: Pozi-Drive: (Box of 200) |
| 372092 | Wood Screw: No8 x 1-1/2": Steel Countersunk: Pozi-Drive. (Box of 200)  |
| 372069 | Wood Screw: No8 x 3/4": Countersunk: (Box of 200)                      |